

RUSHI ASHOK SAVANI

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SUMMARY

Applied AI and Machine Learning Engineer with end-to-end experience designing and deploying deep learning and natural language processing (NLP) solutions on Amazon Web Services (AWS). Proven record building Generative AI applications such as RAG systems, intelligent search, chatbots, and autonomous agents in regulated production environments. Collaborates with cross-functional teams to identify AI integration opportunities aligned with business objectives, iterating through direct testing and evaluation.

WORK EXPERIENCE

AI/ML Engineer Intern

Jun 2025 – May 2026

ProHealth Group

Atlanta, GA

- Designed and deployed an agentic RAG workflow on AWS Bedrock, chaining intent classification, schema-aware retrieval, and Claude SQL generation with Titan embeddings and OpenSearch Serverless, returning business insights under 10 seconds.
- Implemented deep learning and natural language processing (NLP) pipelines on Vertex AI and Gemini, surfacing preventable financial and operational anomalies worth an estimated \$100K per year within Financial Services-adjacent operations.
- Trained an XGBoost rehospitalization classifier using L1 regularization, VIF selection, chi-square filtering, and SMOTE-Tomek balancing, lifting precision from 64% to 82% on clinical and claims data.
- Built LLM evaluation pipelines on 25-sample golden sets scoring factual grounding, citation correctness, and refusal on out-of-scope clinical queries, catching prompt regressions before staging and guiding Claude versus Gemini selection.
- Productionized ML and LLM services as Dockerized FastAPI endpoints on Terraform-managed AWS (ECS, S3, IAM, Secrets Manager), GitHub Actions CI/CD promoting versioned releases across dev and prod, isolating model and prompt change.

Research Assistant

Jan 2025 – May 2025

Syracuse University

Syracuse, NY

- Fine-tuned a RoBERTa sentiment classifier in HuggingFace Transformers on 5,000+ labeled AI-adoption survey responses, tracking experiments and hyperparameters in MLflow with stratified validation and weighted evaluation to reach 91% accuracy.
- Automated survey data cleaning, statistical summaries, and Power BI dashboard generation in Python with pandas and scipy, cutting the research team's reporting cycle from two weeks to three days.

Data Analyst

Jun 2023 - Jun 2024

Krishna Jewels

Mumbai, IN

- Designed and implemented a relational MySQL data model spanning inventory, sales, supplier, and accounting domains, migrating 40K+ records into a queryable schema and reducing lookup time by 70%.
- Engineered automated ETL pipelines feeding Power BI dashboards with daily scheduled refreshes, eliminating 15+ hours/week of manual reporting and giving stakeholders self-serve access to transactional data.

TECHNICAL SKILLS

- Programming & Backend:** Python, SQL, TypeScript, FastAPI, REST APIs
- Machine Learning:** PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers, MLflow, Fine-tuning, Model Deployment
- GenAI & LLM Systems:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Agentic Workflows, Prompt Engineering, LLM Evaluation, Vector Databases (Qdrant, OpenSearch), AWS Bedrock, Vertex AI, Gemini, Claude
- Cloud & MLOps:** AWS (IAM, ECS, S3, Secrets Manager), Microsoft Azure, Terraform, Docker, GitHub Actions CI/CD
- Data Engineering:** PostgreSQL, MySQL, Snowflake, DBT, ETL Pipelines, Power BI, Tableau

PROJECTS

PrivateRAG | Private On-premises Q&A System

- Optimized retrieval accuracy for long clinical documents by layering BM25 sparse matching with dense vector search via Reciprocal Rank Fusion and reranking top candidates through a cross-encoder, cutting p50 latency by 68% on a 3B quantized local model and eliminating ~\$140/month in cloud LLM costs.
- Deployed a zero-egress RAG stack with quantized Llama 3.1, nomic-embed-text embeddings, and self-hosted Qdrant vector search, exposed through a TypeScript UI over a FastAPI backend for regulated healthcare Q&A.

KarmaScore Predictor | Subscription Churn Analysis

- Predicted subscriber churn across 10K+ records using XGBoost, reducing 60 raw features to 23 through importance ranking and correlation filtering to achieve 89% precision on the churn class.
- Translated SHAP feature attributions into a profit-weighted intervention score ranking at-risk subscribers by retention value rather than raw churn probability, exposed through an interactive Streamlit dashboard for prioritized outreach.

JobFlow AI | LLM-Powered Assistant

- Launched a multi-user AI job-application assistant on Mistral and Gemini handling resume tailoring, cover-letter generation, requirement extraction, and fit scoring, scaling daily output from 8 manual to 20 personalized submissions.
- Released the assistant as containerized API microservices on Vercel (PostgreSQL, S3, Google Auth, RBAC-enforced multi-user access), with A/B-tested prompt loops feeding user feedback into production for tailoring and fit scoring.

EDUCATION

Master of Science | Information Systems, Data Science

Syracuse University, School of Information Studies, Syracuse, NY

May 2026

(GPA: 3.85/4)

Bachelor of Engineering | Information Technology

Thadomal Shahani Engineering College, India

May 2024

(GPA: 8.52/10)